Lab 06: System Troubleshooting

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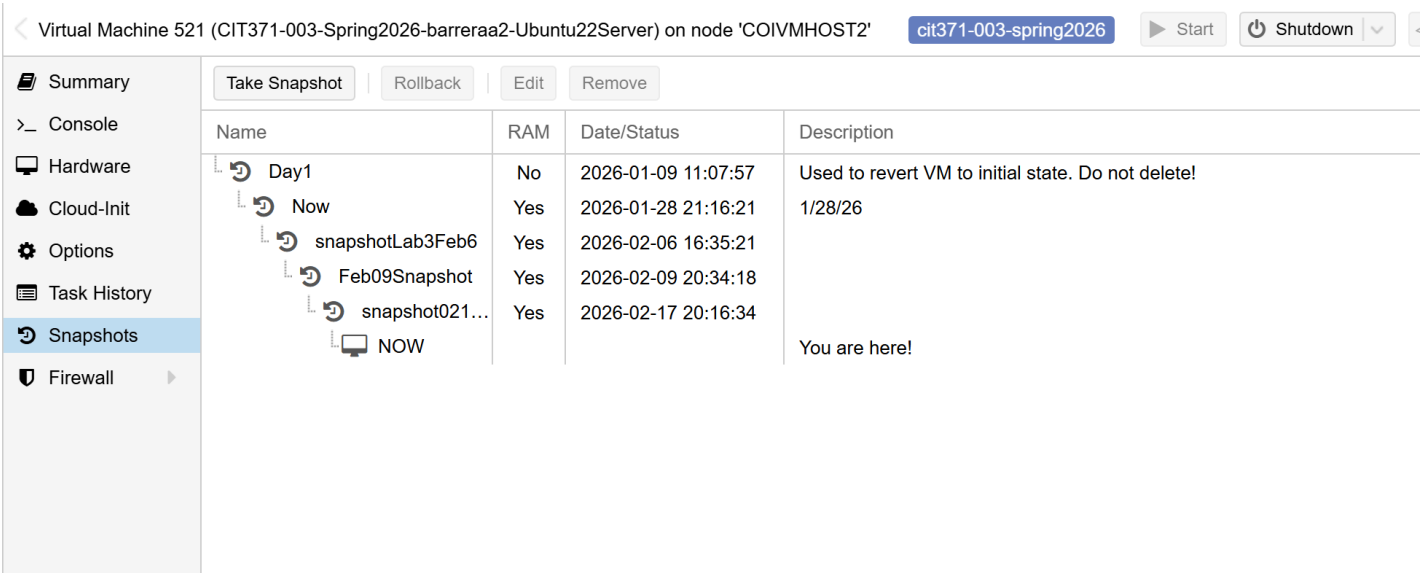
VM Snapshot(s)



Overview

Part 2: Hardware Troubleshooting

In this part we will first find linux commands that gives us the following information on the ubuntu server.

We will start on finding a details list of the VMs hardware configuration. We should get a good amount of information but we will get a pipe to give us only the fist twenty lines.

With this command we will get what we need on the hardware configuration:

```
sudo lshw | head -n 20
```

- On this other request we will now ask for the list of the VM's CPUs information with this command below:

```
lscpu
```

- For this next part we will need to show the VM's memory statistics, this needs to print only the metrics of the VM's total, used, free, shared, cached, and available memory in a table-like format. With this command we will get this type of information:

```
free -h
```

- Now we will give a list of information on the VM's PCI buses with this command:

```
lspci
```

PART 3 : Filesystem troubleshooting

In this part we will get information from the Ubuntu server, first we need a list of all block devices on the VM, this should be using an option to exclude all the loo block devices. This command will list all the block devices, then we will print their information while excluding loop block devices with this commands - we will excluded the loop devices so we will use the -e option and loop devices major number would be 7.

```
sudo lsblk -e 7
```

- In this part - we will now get a list of all the VM's available disk space per filesystem - this print should be human-readable format. With this command we will get that information:

```
df -h
```

For the last part on this section we will now get the list of the partitions on the disk /dev/vda with this command:

```
sudo lsblk /dev/vda
```

Part 4: Performance Troubleshooting

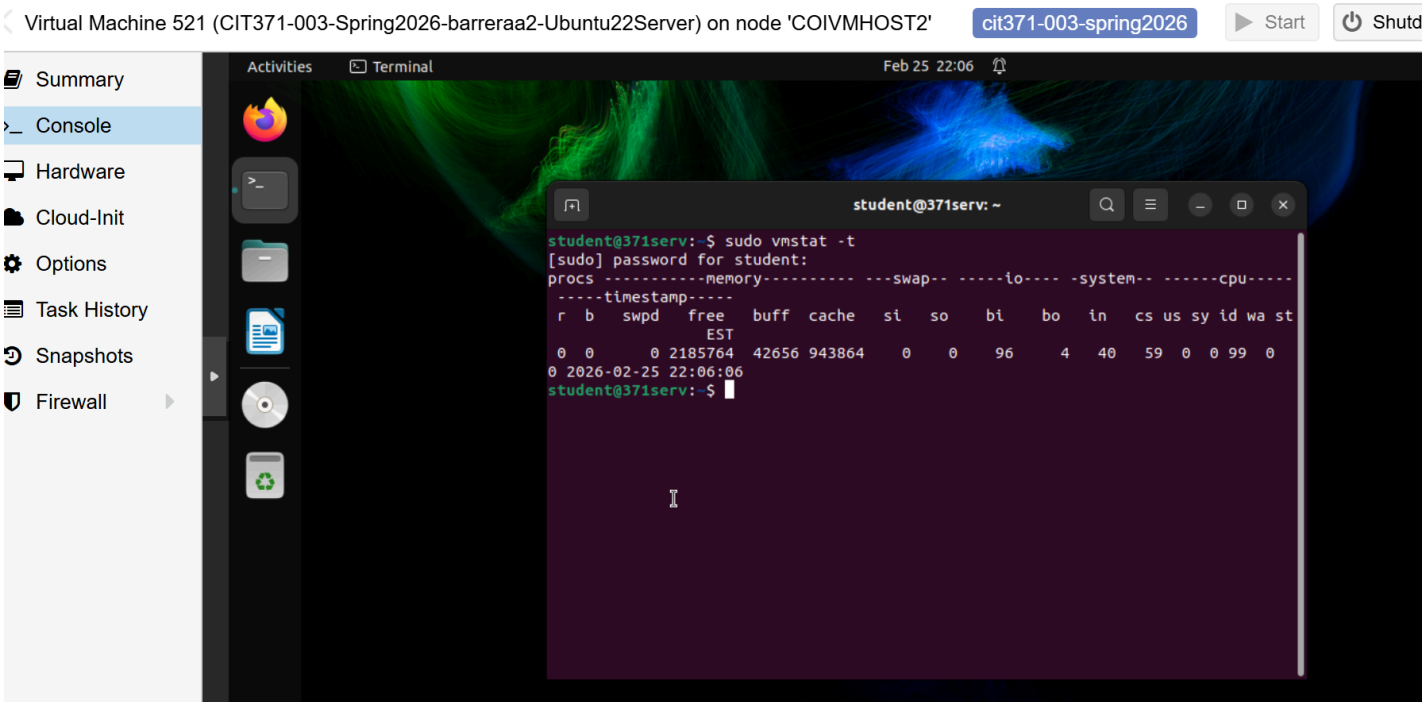
In this part we will start seeing the performance of the Ubuntu Server and troubleshooting any issues related.

First we will get a displace of the active processes - While this command is running we will press “q” to make sure it stops.

```
sudo top
```

For the next part - We will need to give a report on the virtual memory statistics with the timestamps: With this command we will give that printed: With the -t option.

```
sudo vmstat -t
```



Lastly , we will need report on the disk input and outputs statistics - This is a tool call 'iostat', which is part of the ‘sysstat’ package that we must install first.

```
sudo apt install sysstat
```

After we install it with this command, we will need now to run it with this other command:

```
sudo iostat
```

After we have run it, to get more detailed per-device Input and output we will use this command:

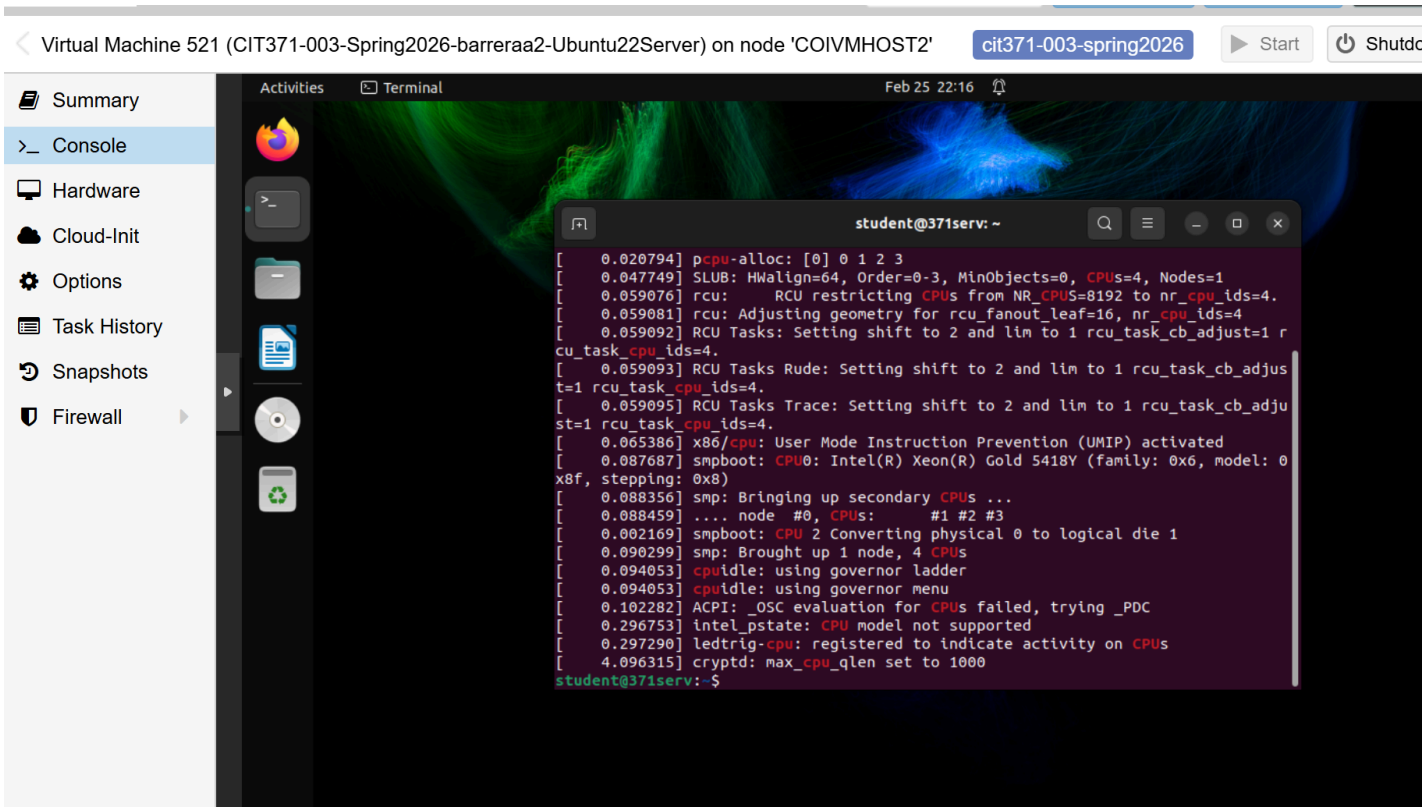
```
sudo iostat -dx
```

PART 5: Miscellaneous Troubleshooting

For this part - We will do random troubleshooting and we will start first with Kernel commands, to show messages, logged-in users and open files:

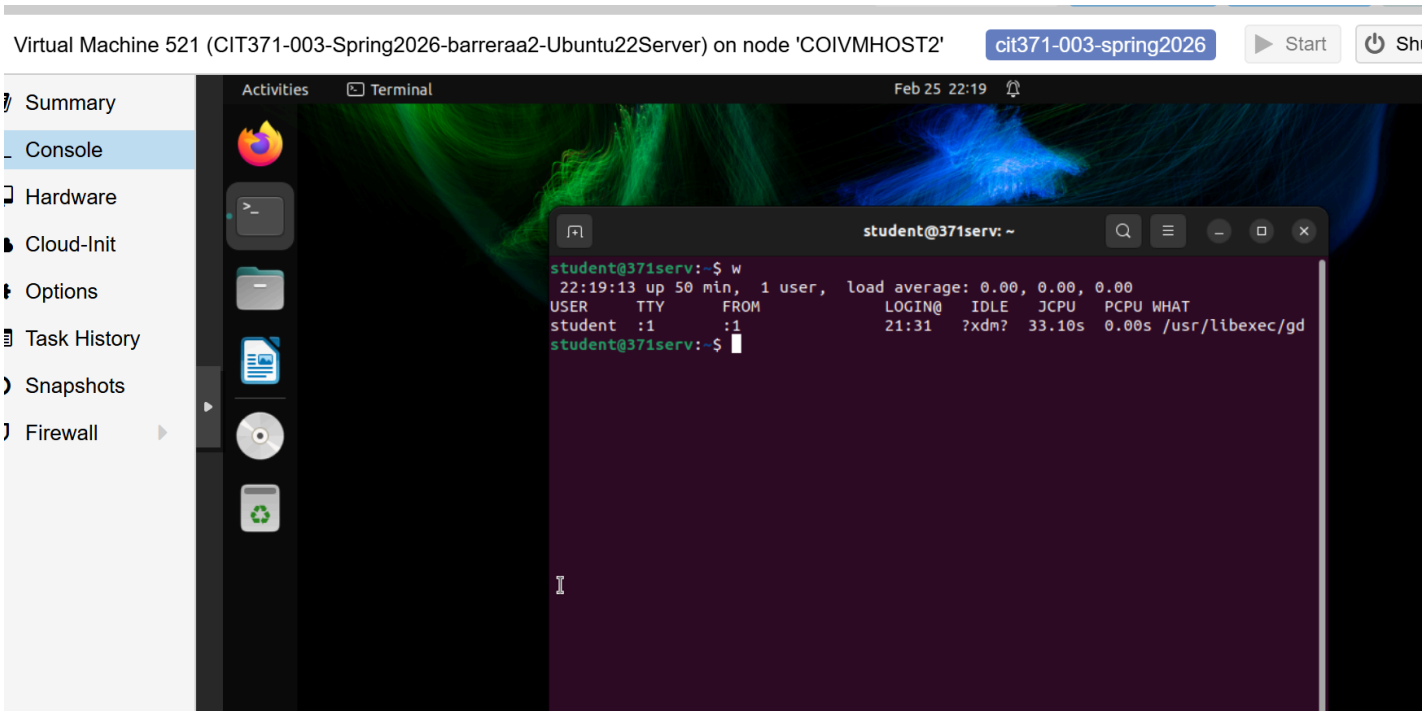
With this command we will view the kernel ring buffer and then we will use the pipe to grep:

```
sudo dmesg | grep -i cpu
```



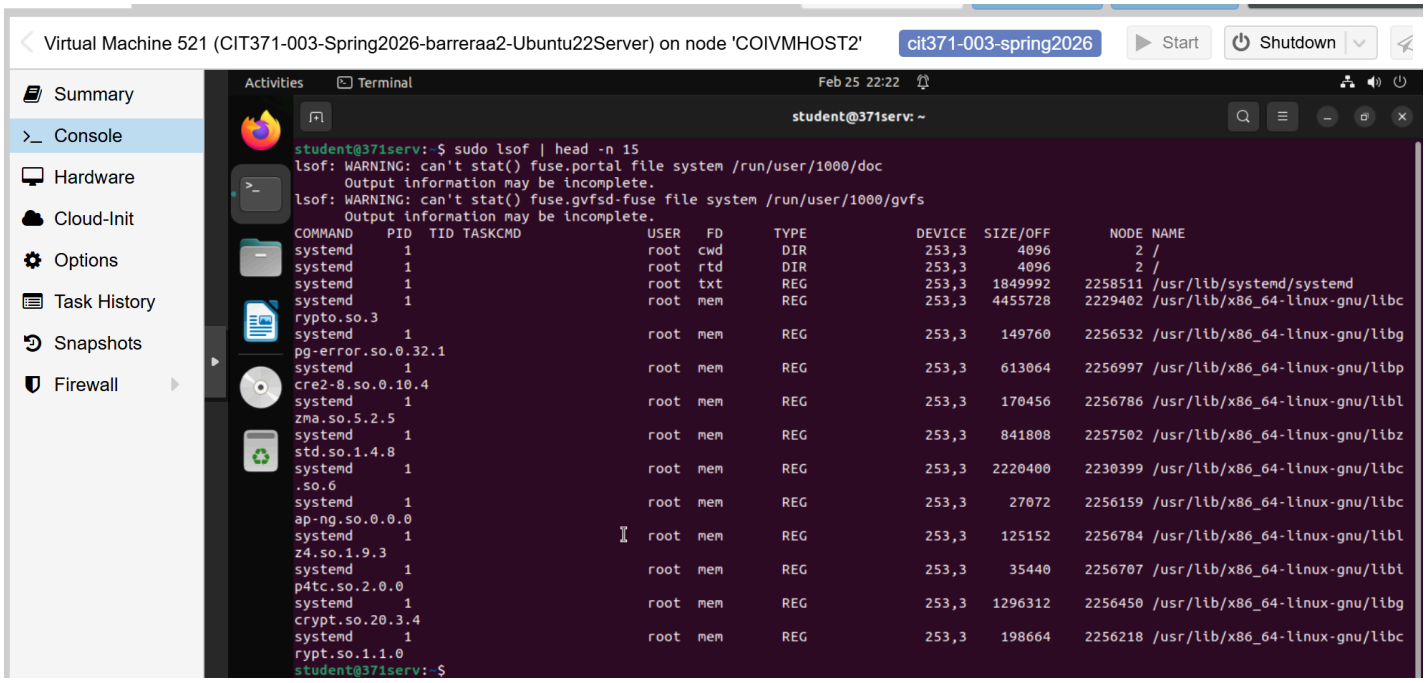
Then we will get a list of the logged-in users and their activities - With this command we will check the user, their TTY, Login time, idle time and what are they running in the CL:

```
w
```



Lastly, we will show any open files on the VM, this will be a list of only 15 lines that will show of files using this command:

```
sudo lsof | head -n 15
```



Reference

Google.com

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